

Two dipole kernels, integrated over the emission point

$$\int d^2 z \frac{(z - w') \cdot (z - w)}{(z - w')^2 (z - w)^2} \frac{(x - z) \cdot (y - z)}{(x - z)^2 (y - z)^2}$$

Notation: $\mathcal{K}(u, v) \equiv \frac{u \cdot v}{u^2 v^2}$, so the integrand is $\mathcal{K}(z - w', z - w) \mathcal{K}(x - z, y - z)$

0. The answer, and the answer to “is it divergent”

It converges. There is nothing to regulate. The integral is finite at all four coincidence points and at infinity, for any four distinct w', w, x, y , and it has a closed form:

$$\int d^2 z \mathcal{K}(z - w', z - w) \mathcal{K}(x - z, y - z) = -\frac{\pi}{2} \left[\mathcal{K}(w' - x, w - y) \log \mathcal{X}_1 + \mathcal{K}(w' - y, w - x) \log \mathcal{X}_2 \right]$$

$$\mathcal{X}_1 = \frac{(w - w')^2 (x - y)^2}{(w' - y)^2 (w - x)^2}, \quad \mathcal{X}_2 = \frac{(w - w')^2 (x - y)^2}{(w' - x)^2 (w - y)^2}.$$

The two arguments are **conformal cross-ratios** — dimensionless, as they must be, since the integral has mass dimension $-2 + 1 + 1 = 0$ and no scale of its own.

1. Why nothing diverges

Short distance. Each factor is a *dipole* kernel, and a dipole kernel is only $O(1/\rho)$ at its endpoints, not $O(1/\rho^2)$: as $z \rightarrow w'$,

$$\frac{(z - w') \cdot (z - w)}{(z - w')^2 (z - w)^2} \rightarrow \frac{(z - w') \cdot (w' - w)}{(z - w')^2 (w' - w)^2} = O\left(\frac{1}{\rho}\right),$$

because the numerator vanishes linearly. And **no point is an endpoint of both kernels** — w', w belong to the first, x, y to the second. So at each of the four singular points only one kernel is singular, the integrand is $O(1/\rho)$, and $\int \rho d\rho \rho^{-1}$ converges. Measured $\rho \times \langle |\text{integrand}| \rangle_\theta$, constant to five digits over two decades:

point	$\rho = 10^{-2}$	$\rho = 10^{-4}$
$z \rightarrow w'$	0.200877	0.200867
$z \rightarrow w$	0.071781	0.071779
$z \rightarrow x$	0.067050	0.067052
$z \rightarrow y$	0.044924	0.044903

Large distance. $\mathcal{K} \rightarrow 1/z^2$ for either kernel, so the product falls as $1/z^4$ and $\int \rho d\rho \rho^{-4}$ converges. Measured $\rho^4 \langle \text{integrand} \rangle_\theta \rightarrow 1$:

rho=1e+01	rho^4 x <integrand> = +0.988745	rho^2 x <integrand> = 9.887e-03
rho=1e+02	+0.999889	9.999e-05
rho=1e+03	+0.999999	1.000e-06
rho=1e+04	+1.000000	1.000e-08

This is the crucial difference from the rows we did before, where a light-cone vertex supplied a second denominator on the *same* separation. Here the two kernels never share a point, so neither end of the z integration produces a logarithm.

2. Doing the integral: complex coordinates

Write $z = z_1 + iz_2$. A two-vector v^i/v^2 is the complex number $1/\bar{v}$, and $u \cdot v = \text{Re}(u\bar{v})$, so

$$\mathcal{K}(a - z, b - z) = \frac{(a - z) \cdot (b - z)}{(a - z)^2 (b - z)^2} = \text{Re} \frac{1}{(\bar{z} - \bar{a})(z - b)}$$

(checked against the vector form to twelve digits). Hence the product of the two kernels is

$$\frac{1}{4} \left[\frac{1}{(\bar{z} - \bar{w}')(z - w)} + \text{c.c.} \right] \left[\frac{1}{(\bar{z} - \bar{x})(z - y)} + \text{c.c.} \right] = \frac{1}{2} \text{Re}[(a) + (b)],$$

$$(a) = \frac{1}{(\bar{z} - \bar{w}')(\bar{z} - \bar{x})(z - w)(z - y)}, \quad (b) = \frac{1}{(\bar{z} - \bar{w}')(\bar{z} - \bar{y})(z - w)(z - x)},$$

the remaining two cross terms being the complex conjugates of these. Both are the same master,

$$M(a_1, a_2; b_1, b_2) = \int d^2 z \frac{1}{(\bar{z} - \bar{a}_1)(\bar{z} - \bar{a}_2)(z - b_1)(z - b_2)}.$$

The master by partial fractions

$$\frac{1}{(\bar{z} - \bar{a}_1)(\bar{z} - \bar{a}_2)} = \frac{1}{\bar{a}_1 - \bar{a}_2} \left[\frac{1}{\bar{z} - \bar{a}_1} - \frac{1}{\bar{z} - \bar{a}_2} \right],$$

and likewise for the holomorphic pair, so everything reduces to the single logarithm

$$L(a, b) = \int_{|z| < R} d^2 z \frac{1}{(\bar{z} - \bar{a})(z - b)} = \pi \log \frac{R^2}{|a - b|^2}.$$

Derivation. Shift $u = z - a$, $c = b - a$, use $\frac{1}{u} = \partial_{\bar{u}} \log |u|^2$ and $\partial_{\bar{u}} \frac{1}{u - c} = \pi \delta^{(2)}(u - c)$; Stokes ($\int_D d^2 u \partial_{\bar{u}} F = \frac{1}{2i} \oint_{\partial D} du F$) leaves $\pi \log R^2$ from the boundary and $-\pi \log |c|^2$ from the delta. Checked by quadrature: $R = 100$ gives 27.381423 against 27.381004; $R = 500$, 37.493726 against 37.493400.

The cutoff cancels

The four terms come with signs $+, -, -, +$, which sum to zero, so $\log R^2$ drops out and the result is a ratio of the four distances:

$$M(a_1, a_2; b_1, b_2) = \frac{-\pi}{(\bar{a}_1 - \bar{a}_2)(b_1 - b_2)} \log \frac{|a_1 - b_1|^2 |a_2 - b_2|^2}{|a_1 - b_2|^2 |a_2 - b_1|^2}$$

That the answer is finite here is the same statement as the power counting of §1: the $\log R$ that each partial fraction carries is an artefact of splitting a convergent integral.

3. Assembling

Term (a) is $M(w', x; w, y)$ and term (b) is $M(w', y; w, x)$. Taking $\frac{1}{2} \text{Re}$ and using $\text{Re} \frac{1}{uv} = \mathcal{K}(u, v)$ once more (the logarithms are real) turns the prefactors straight back into dipole kernels:

$$\begin{aligned} & \int d^2 z \frac{(z - w') \cdot (z - w)}{(z - w')^2 (z - w)^2} \frac{(x - z) \cdot (y - z)}{(x - z)^2 (y - z)^2} \\ &= -\frac{\pi}{2} \left[\frac{(w' - x) \cdot (w - y)}{(w' - x)^2 (w - y)^2} \log \frac{(w - w')^2 (x - y)^2}{(w' - y)^2 (w - x)^2} + \frac{(w' - y) \cdot (w - x)}{(w' - y)^2 (w - x)^2} \log \frac{(w - w')^2 (x - y)^2}{(w' - x)^2 (w - y)^2} \right] \end{aligned}$$

The structure is one term per way of pairing $\{w', w\}$ against $\{x, y\}$; in each, the logarithm's denominator uses the *other* pairing.

Checks

Symmetry. The integrand is separately even under $w \leftrightarrow w'$, under $x \leftrightarrow y$, and under $(w, w') \leftrightarrow (x, y)$. Each of those swaps exchanges the two terms of the answer, so the total is invariant — numerically identical to twelve digits:

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base      +0.396584299678
w <-> w'  +0.396584299678
x <-> y   +0.396584299678
both      +0.396584299678
(w, w') <-> (x, y) +0.396584299678
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Against direct quadrature (partition of unity around the four singular points, polar about each):

trial 0: quad +1.495565701	closed form +1.495566060	rel 2.40e-07
trial 1: quad +2.939965670	closed form +2.939965321	rel 1.19e-07
trial 2: quad +0.508794749	closed form +0.508786643	rel 1.59e-05
trial 3: quad +1.185187569	closed form +1.185186898	rel 5.65e-07

4. Where a divergence *does* appear

Not in z — in the *external* points, and only when two of them collide. The formula develops a logarithm at each of the six pairings, and each has a clean origin:

- $w \rightarrow w'$ or $x \rightarrow y$ — one kernel degenerates, $\mathcal{K}(z-w, z-w) = \frac{1}{(z-w)^2}$, and the z integral really is log divergent there. It shows up as the $\log(w-w')^2$ and $\log(x-y)^2$ in the numerators of $\mathcal{X}_1, \mathcal{X}_2$.
- $w' \rightarrow x, w' \rightarrow y, w \rightarrow x, w \rightarrow y$ — a point of the first kernel meets a point of the second, so two kernels become singular at the same z and the integrand is $O(1/\rho^2)$. It shows up in the denominators. Numerically, with $|x-w'| \rightarrow 0$ the result grows as $-\frac{\pi}{2}\mathcal{K}(w'-y, w-x) \log \frac{1}{|x-w'|^2}$ (ratio 0.4588, 0.4640, 0.4666 at $|x-w'| = 10^{-4}, 10^{-6}, 10^{-8}$, creeping toward the predicted 0.47436 as the subleading constant is divided out).

All six are **logarithms against a two-dimensional measure**, so they are integrable in the remaining $\int_{x,y,w,w'}$ and need no regulator either. In particular $\int d^2(w-w') e^{-ik \cdot (w-w')} \log(w-w')^2$ is a perfectly good distribution.

5. Back into the cross section

With $\frac{1}{(2\pi)^3} \frac{q^4}{8\pi^3} \frac{1}{2\pi^2} \times \left(-\frac{\pi}{2}\right) = -\frac{q^4}{256\pi^7} = -\frac{\alpha_s^2}{16\pi^5}$,

$$\begin{aligned} \frac{d^3 N_{\text{NLO}}}{d^3 k} &= -\frac{\alpha_s^2 N_c}{16\pi^5 k^+} \log \frac{\vee}{\Lambda} \int_{x,y,w,w'} e^{-ik \cdot (w'-w)} \rho^c(x) \rho^c(y) \\ &\times \left[\frac{(w'-x) \cdot (w-y)}{(w'-x)^2 (w-y)^2} \log \frac{(w-w')^2 (x-y)^2}{(w'-y)^2 (w-x)^2} + \frac{(w'-y) \cdot (w-x)}{(w'-y)^2 (w-x)^2} \log \frac{(w-w')^2 (x-y)^2}{(w'-x)^2 (w-y)^2} \right]. \end{aligned}$$

The only divergence left in the expression is the one that was already there: $\log \frac{\vee}{\Lambda}$, the rapidity logarithm from the p^+ integration. It is not a transverse divergence and it is not removed by a transverse regulator — it is absorbed into the rapidity evolution of the source correlator $\rho^c(x) \rho^c(y)$.

zint/src/twokernel.py — the closed form, the master, and the quadrature; zint/src/tk_fast.py — symmetries, power counting, coincidence limits. Output in zint/RESULTS_twokernel.txt.